# How much of a lap time is the car

Formula 1, 2023. 128 teammate comparisons from 20 races, with every lap spent stuck behind another car held out.

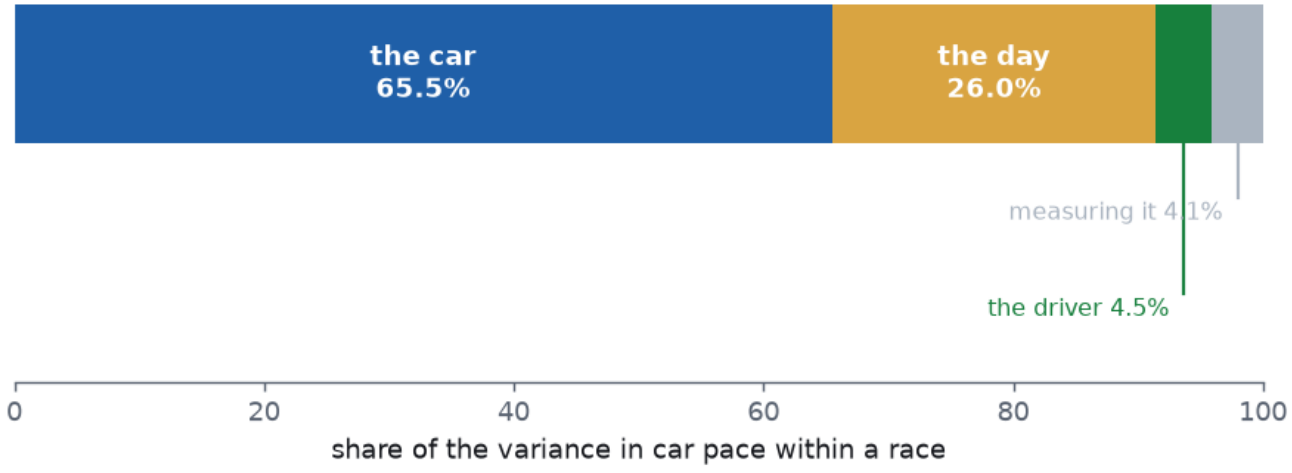
## The car is 3.8 times the driver

0.732 s a lap between teams against 0.192 s a lap between teammates — 15 times in variance. And what happens to a car on a given afternoon is 2.4 times the driver: the term nobody ranks.

| 0.732 | 0.192 | 3.8× | 2 of 12 |
|---|---|---|---|
| the car, in seconds a lap | the driver | car over driver | pairings that survive |

A driver is only measurable against someone who had the same car. Each driver's pace is fitted on their own clear-air laps with tyre age and fuel load taken out, and the difference between two teammates on one afternoon is one comparison.

**The driver is four and a half per cent of it**
car 0.73 s a lap, the day 0.46, the driver 0.19



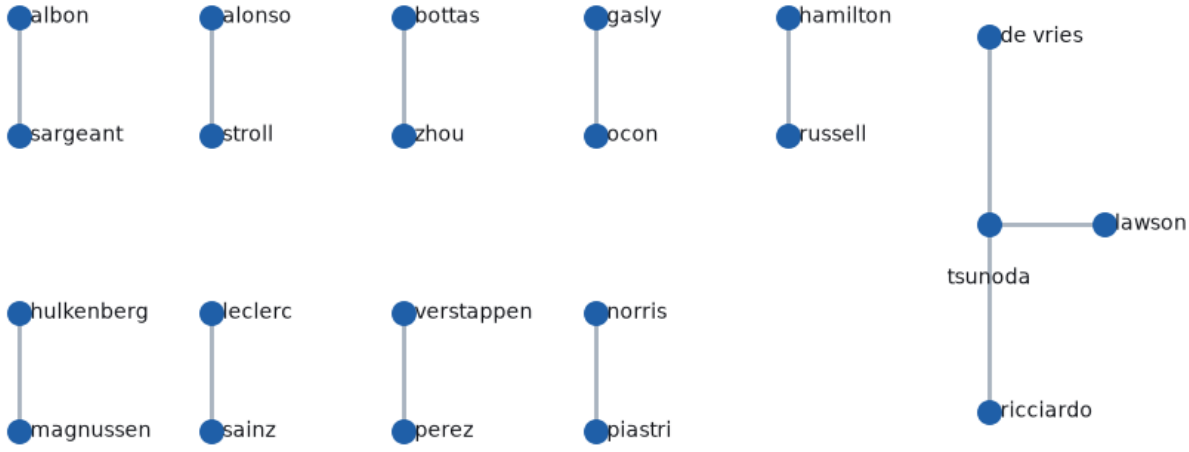
# First: what can be compared with what

Drivers are nodes, a shared team-race is an edge, and the shape of that graph decides which questions have answers before any statistic is computed.

The graph is a forest. A forest on 22 nodes with 10 components has exactly 12 edges, and there are 12 — so 0 cycles. Every comparison rests on a single pairing with no second route to check it against, and 10 of the 22 driver effects are free parameters this season cannot pin down at all. There is no chain of shared cars from Verstappen to Hamilton, and no statistic can manufacture one.

## Twenty-two drivers, ten islands

an edge is a shared car; 0 cycles, so no comparison has a second route to check it against




22 driver effects on a design of rank 12: 10 of them are free parameters no amount of 2023 data can fix.

| property | count |
|---|---|
| drivers | 22 |
| pairings | 12 |
| components | 10 |
| identified | 12 |
| free | 10 |
| cycles | 0 |
| largest component | 4 |
| isolated pairs | 9 |

# Who can be compared with whom

| size | drivers who can be compared |
|---|---|
| 4 | de_vries, lawson, ricciardo, tsunoda |
| 2 | albon, sargeant |
| 2 | alonso, stroll |
| 2 | bottas, zhou |
| 2 | gasly, ocon |
| 2 | hamilton, russell |
| 2 | hulkenberg, kevin_magnussen |
| 2 | leclerc, sainz |
| 2 | max_verstappen, perez |
| 2 | norris, piastri |

## The comparisons themselves

One car, two drivers, one afternoon.

Laps within two seconds of the car ahead are held out, because being held up costs about a quarter of a second a lap and is a property of the traffic rather than the driver.

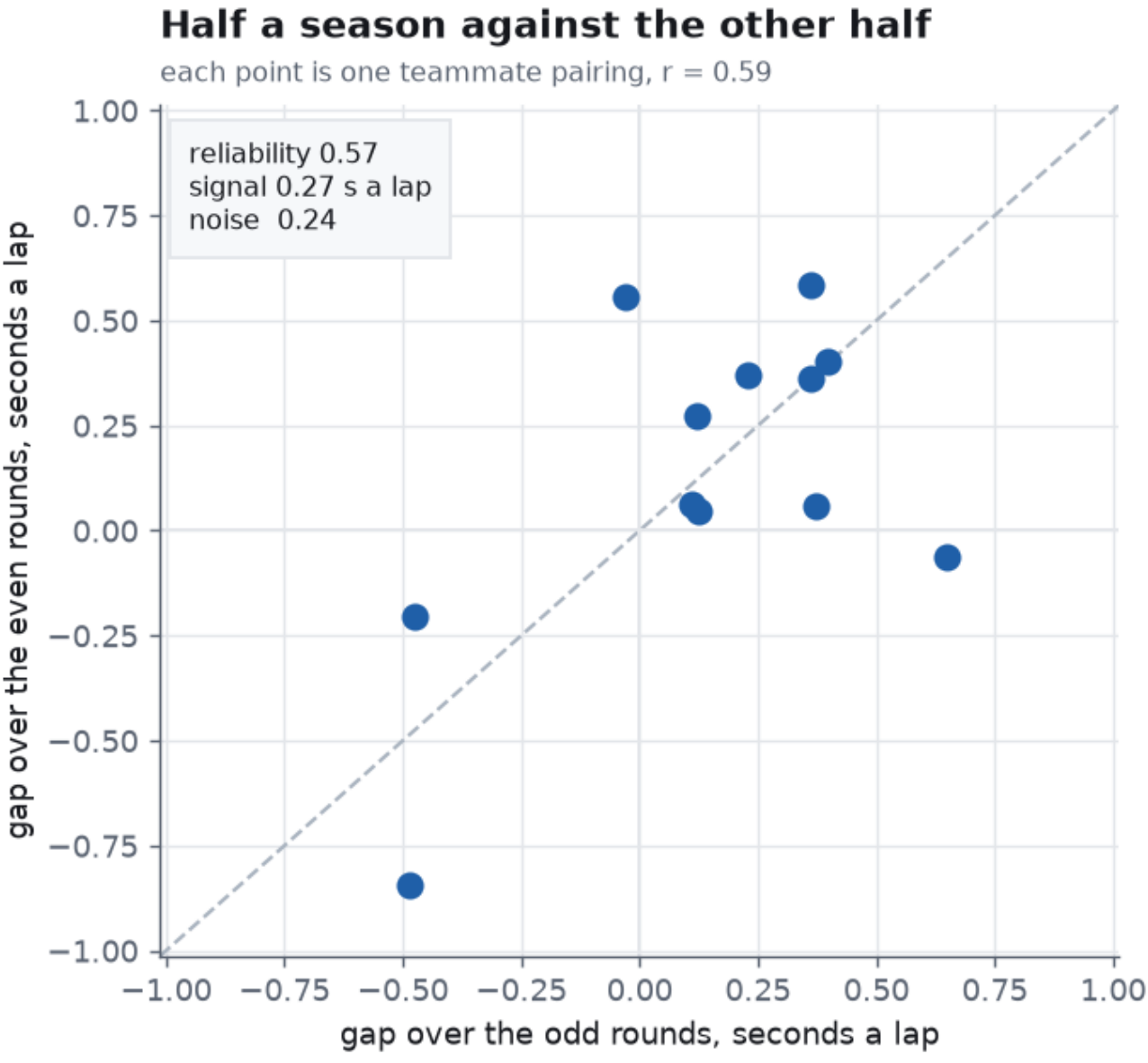
| step | laps or races |
|---|---|
| clean laps | 18,769 |
| held out: in traffic | 8,193 |
| team-races seen | 199 |
| kept: contrasts | 128 |
| refused: too few clear laps from one of them | 40 |
| refused: only one driver classified | 21 |
| refused: age and lap not separable | 9 |
| refused: a gap too large to be pace | 1 |

## Splitting the spread

Within a race, the spread of car paces splits into a part between teams (0.732 s a lap) and a part within a team (0.531 s a lap, over 178 team-races).

The within-team part is not the driver. It is the driver plus damage, a strategy call, a lap of traffic the detector missed, and the ordinary error in fitting a pace to forty laps.

The same teammate gap measured on odd rounds against even rounds correlates 0.590 over 12 pairings, a reliability of 0.565. Barely more than half of what separates one pairing from another comes back in the other half of the season. That fraction is what tells the persistent part from the day, and a car's pace in one race is itself only known to ±0.182 s a lap, which is taken out separately.



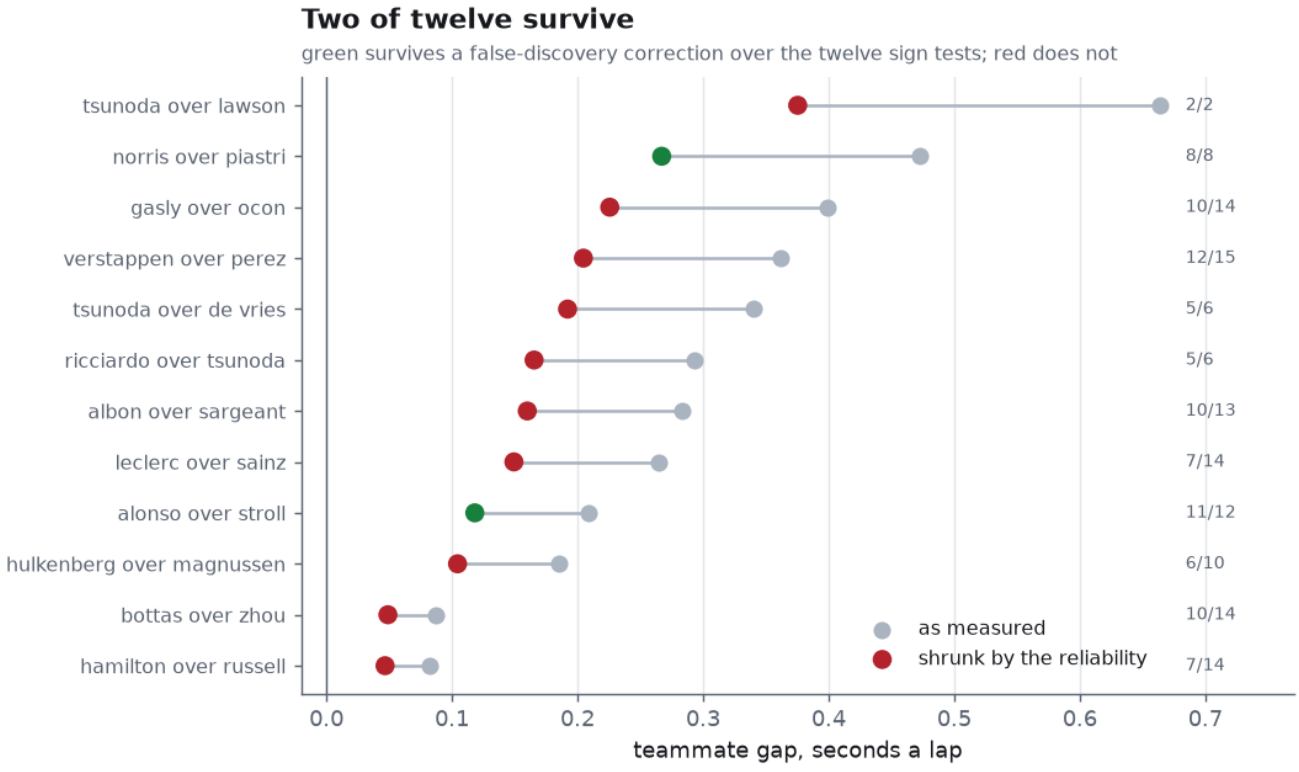
# Splitting the spread (continued)

| what it is | s a lap | share of variance |
|---|---|---|
| the car the team built | 0.732 | 65.5% |
| what happened on the day | 0.461 | 26.0% |
| the driver | 0.192 | 4.5% |
| error in measuring it | 0.182 | 4.1% |

# Every pairing the season supports

The sign test asks only how often the quicker driver was quicker, which needs no model to believe. Shrinking by the reliability pulls every gap toward zero, because a gap over a dozen races is partly those dozen races.

4 pairings reach p<0.05, but 12 tests expect some of that. Controlling the false discovery rate leaves 2: norris over piastri (p 0.0039); alonso over stroll (p 0.0032). The rest of the table is a table, not a set of findings.

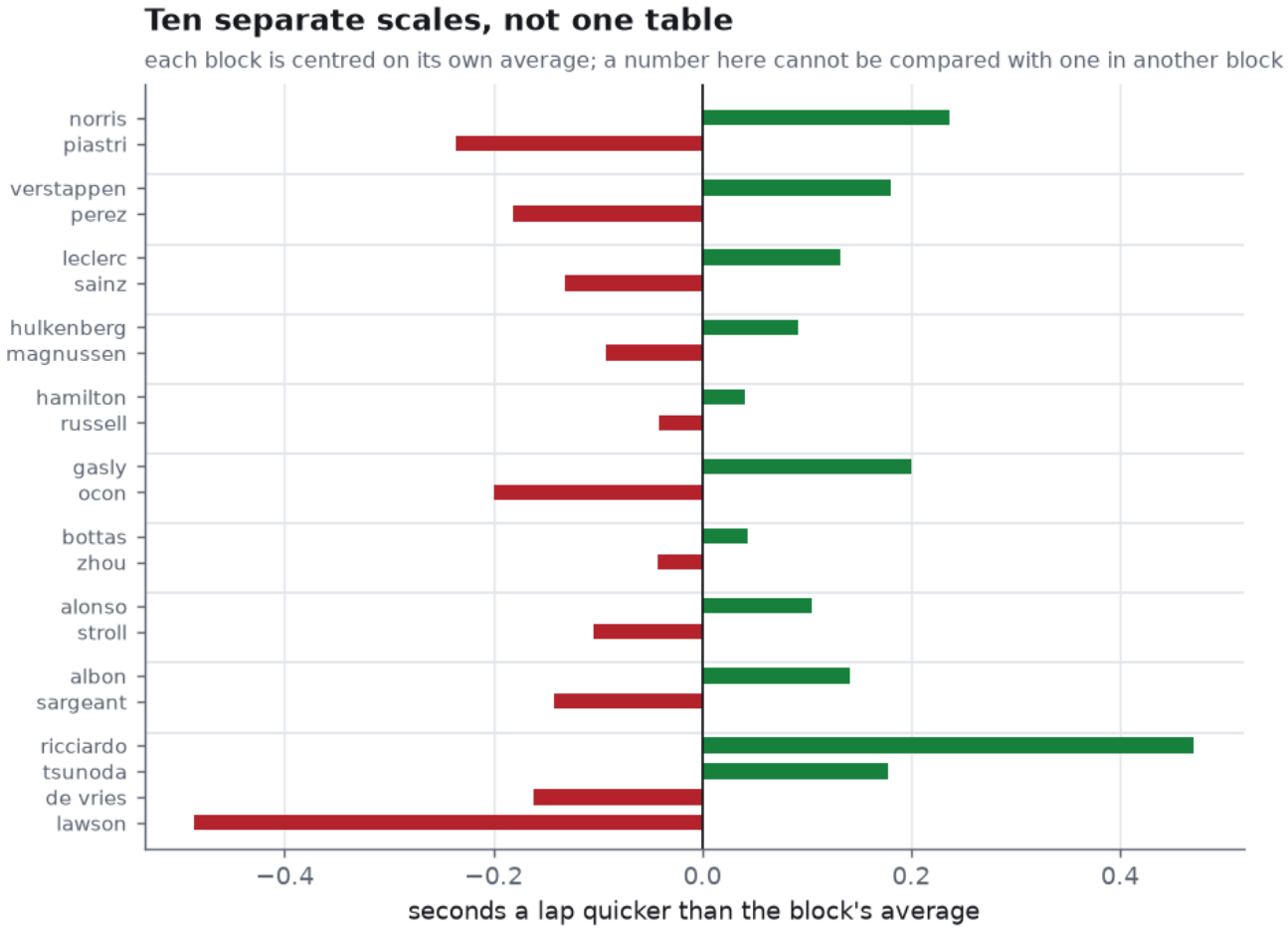



| quicker | slower | raw | shrunk | ahead | p | survives |
|---|---|---|---|---|---|---|
| tsunoda | lawson | +0.664 | +0.375 | 2/2 | 0.2500 | no |
| norris | piastri | +0.473 | +0.267 | 8/8 | 0.0039 | yes |
| gasly | ocon | +0.399 | +0.226 | 10/14 | 0.0898 | no |
| max_verstappen | perez | +0.362 | +0.205 | 12/15 | 0.0176 | no |
| tsunoda | de_vries | +0.340 | +0.192 | 5/6 | 0.1094 | no |
| ricciardo | tsunoda | +0.293 | +0.165 | 5/6 | 0.1094 | no |
| albon | sargeant | +0.283 | +0.160 | 10/13 | 0.0461 | no |
| leclerc | sainz | +0.264 | +0.149 | 7/14 | 0.6047 | no |
| alonso | stroll | +0.209 | +0.118 | 11/12 | 0.0032 | yes |
| hulkenberg | kevin_magnussen | +0.185 | +0.104 | 6/10 | 0.3770 | no |
| bottas | zhou | +0.087 | +0.049 | 10/14 | 0.0898 | no |
| hamilton | russell | +0.083 | +0.047 | 7/14 | 0.6047 | no |

# Driver effects, and why they do not form one list

Seconds a lap quicker than the driver's own component average. Positive is quicker, which is the opposite sign to a lap time.

> Comparing a number here with one from another block is meaningless. Each component is centred on its own mean, and nothing in the season ties one block's zero to another's. A single table ordering all 22 drivers would be inventing 10 numbers the data does not contain.

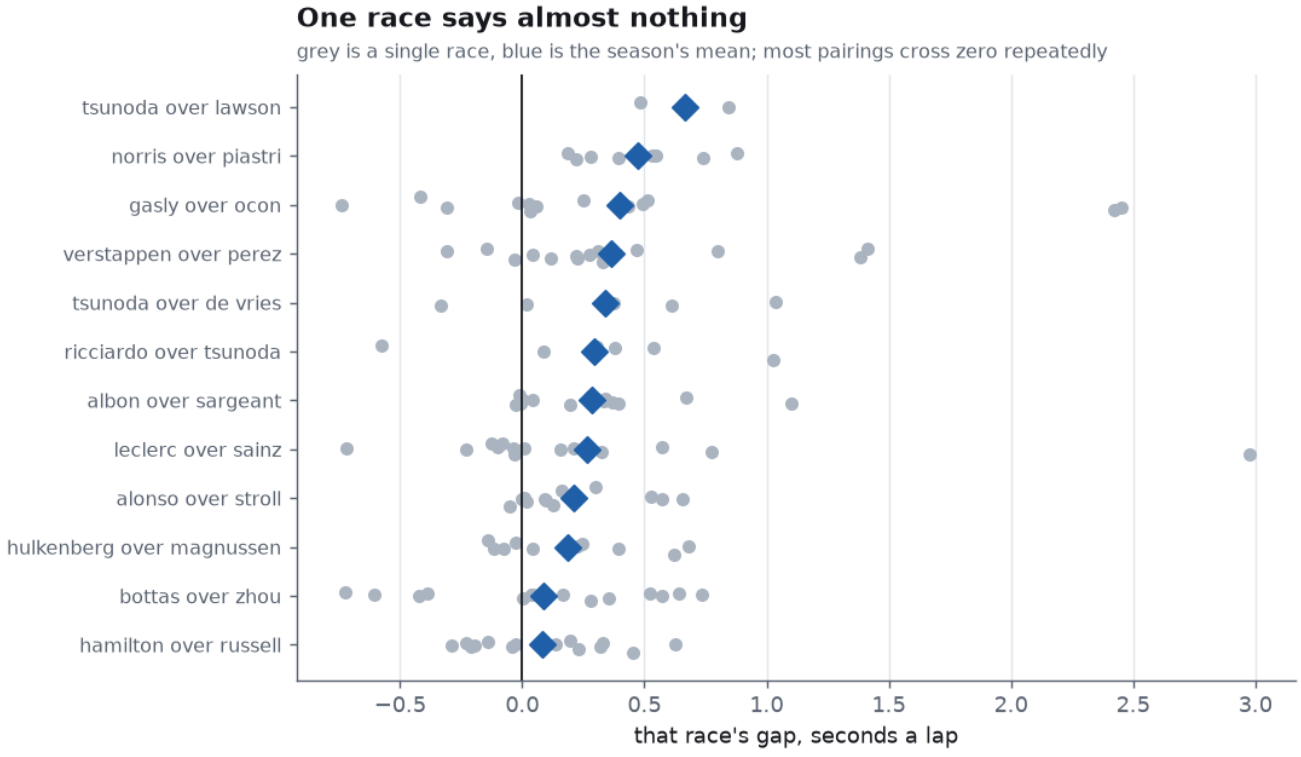



| driver | s a lap |
|---|---|
| ricciardo | +0.471 |
| tsunoda | +0.178 |
| de_vries | -0.162 |
| lawson | -0.486 |

Only the components with three or more drivers are shown, because in a pair the two numbers are the same comparison written twice.

## What this does not do

It does not rank drivers across teams, measure anything in qualifying, or separate a driver from the car's suitability to their style — a car built around one driver flatters that driver, and that appears here as driver skill. It uses one season, which is why the race-to-race term is as large as it is, and it has no tyre compound, because the source does not record one.




Lap times, pit stops and results from the Jolpica mirror of the Ergast schema.